

# Determining the Effect of Trypsin Treatment on Ovalbumin-induced Allergic Reaction in RBL-2H3 Basophilic Cell Lines

## CELL

### Research Question

How does increasing trypsin concentrations affect the severity of basophilic cell degranulation and release of B-Hexosaminadase in response to ovalbumin exposure in egg allergies?

### Background Research

Egg allergies are a prevalent food allergy within children and adults typically caused by an immune reaction to a protein called ovalbumin which is primarily found in egg whites. This protein is the main trigger in allergic reactions. When the protein binds to the IgE antibody in the cell this will lead to the cell to degranulation in which is release granules such as histamine and B-Hexosaminadase which causes the allergic symptoms. According to a recent study, the expression of SERPBINB-10 a gene found in humans is a key factors in allergic reactions(Osborne et al, 2017). The gene is a protease inhibitor in which it blocks the activity of certain protease such as trypsin which has a role in allergic reaction mediation. Overexpression of the gene leading to less trypsin activity which then leads to ovalbumin staying intact in the cell and triggering responses. Investigating the relationship of trypsin activity and severity of allergic reactions can help provide key insight to how SERPINB-10 contributes to allergic reactions.

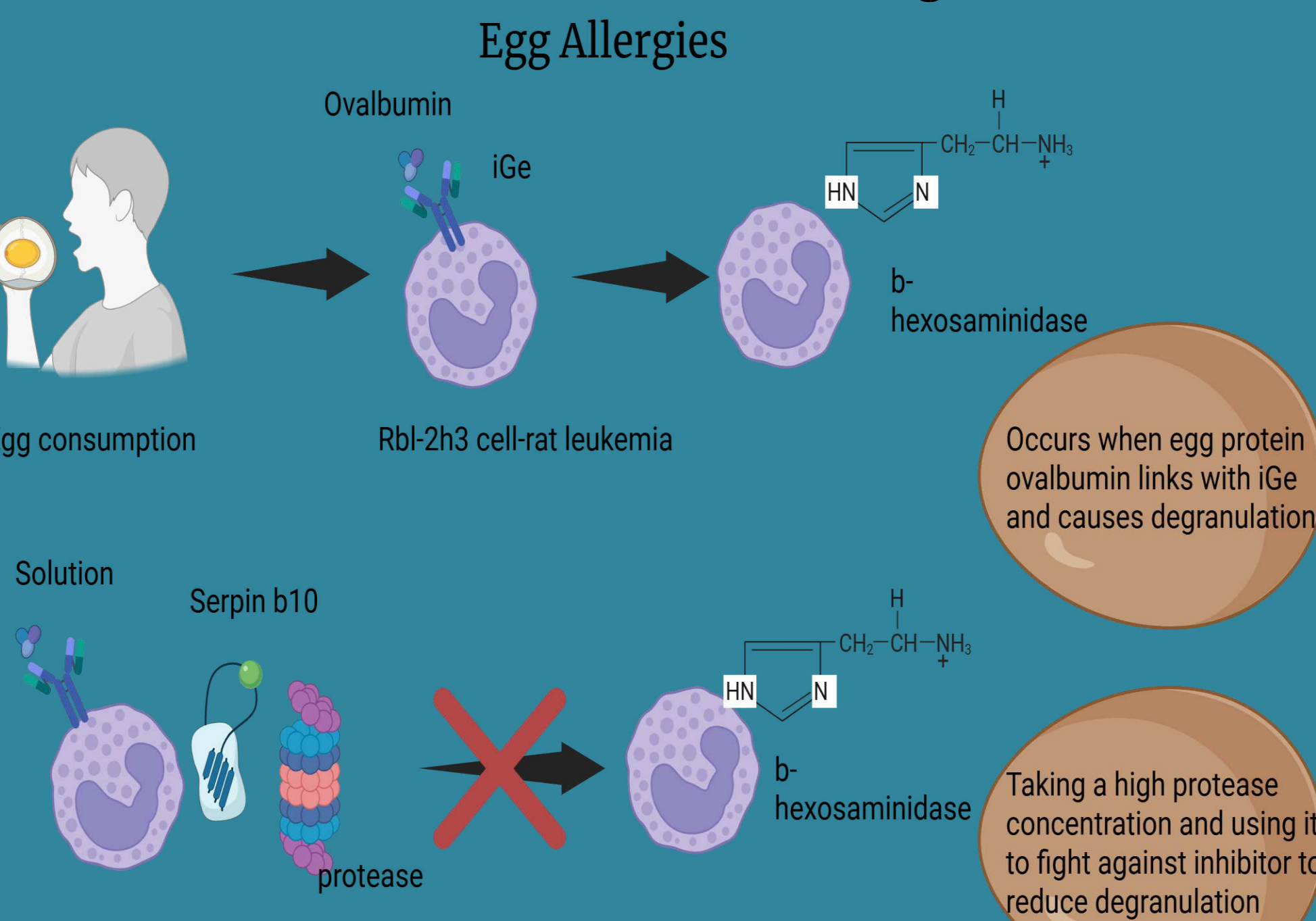


Figure 1: Immunology

### Methodology Expanded

**Independent Variable**  
Trypsin introduced to RBL-2H3 cell line

**Dependent Variable**  
Resulting concentration of B-hexosaminidase released during degranulation

**Control**  
RBL-2H3 cell lines that are left untreated.

### Results

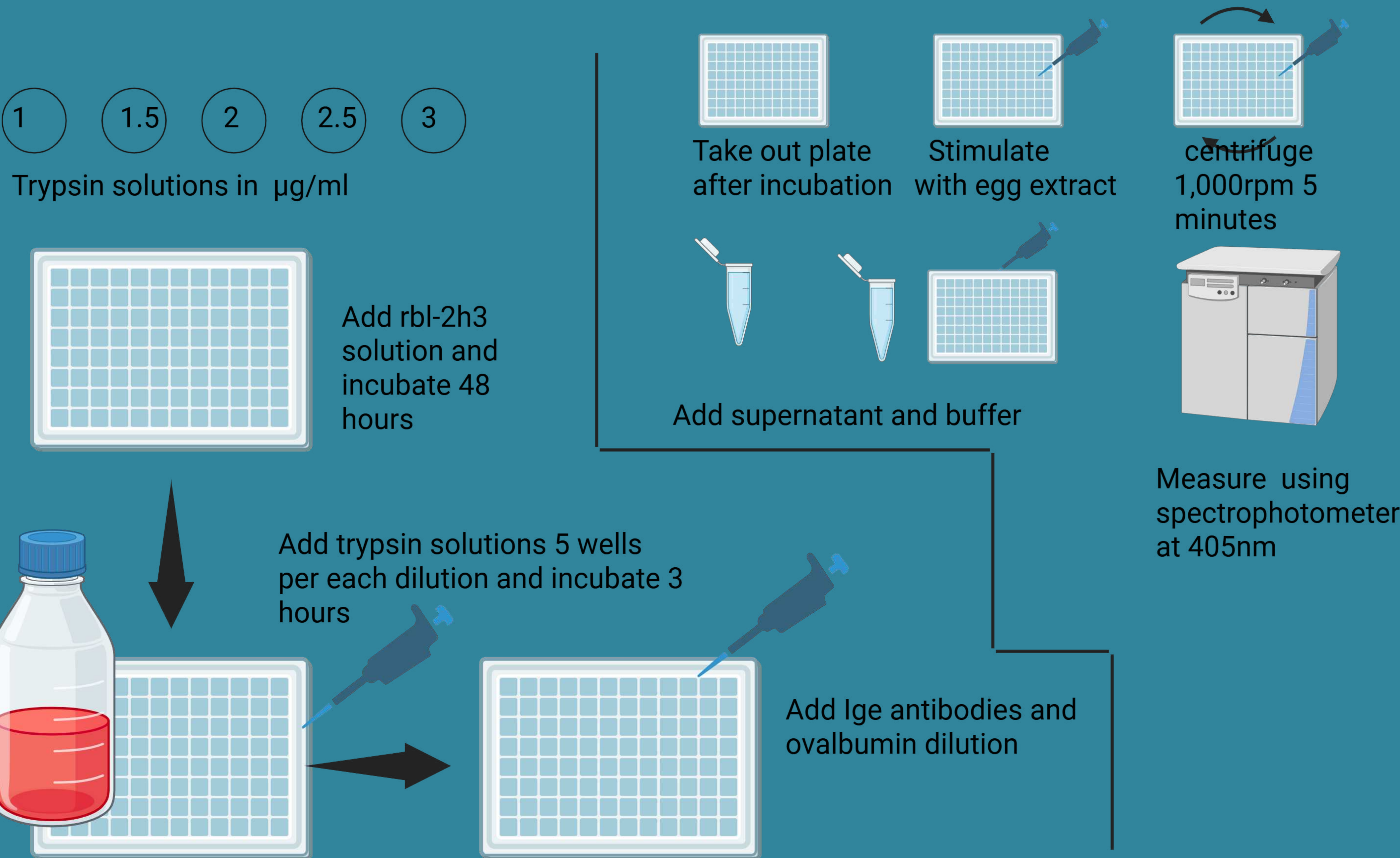
NO DATA

### Conclusion

Data was not be able to be collected this year due to a couple setbacks including time management, acquisition of Tyrode's buffer late, along with multiple solutions having to be made in the lab. 2 RBL-2H3 seeded well plates were able to be treated with trypsin solutions and then sensitized with the IgE antibody. Had there not been issues with creating certain solutions in lab then their would-be data for 3 trials with 5 replicates along with the control wells. For future purposes it is of best interest to get a early head start and stay persistent on completing all steps of the experiment including making solutions and treatments.

### Methodology

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### References

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